CHERI FAVEL

Chemical Engineer (MEng) | Water Treatment & Resource Recovery

Abu Dhabi, UAE | South African | UAE Resident Visa | Availability: Immediate
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Engineering Portfolio | Scan QR to save my contact details



PROFILE

MEng Chemical Engineer with industrial R&D experience in mine-tailings water treatment, dissolved-metal removal and electrochemical resource recovery. Experience includes treatment testwork, analytical data interpretation, mass and energy balances, PFD/P&ID preparation and technical reporting. Additional hands-on support with vessel RO watermakers, pumps and plumbing, supported by environmental engineering training and published mineral-leaching research. Seeking UAE water-treatment, environmental process, applications or R&D opportunities.

CORE COMPETENCIES

- **Industrial water treatment**: mine-tailings water, iron/zinc/manganese removal, oxidation, precipitation, electrochemical treatment, process evaluation and resource recovery.
- **Water chemistry / separation**: pH adjustment, metal speciation considerations, solid–liquid separation, sampling and analysis; understanding of UF/NF/RO principles and fouling/scaling.
- **Process engineering**: mass/energy balances, process calculations, PFDs/P&IDs, equipment-sizing support, design bases, specifications and technical reporting.
- **Laboratory / analytical**: ICP-OES, XRF, XRD, FTIR, SEM-EDS, BET, TGA; experimental design, data interpretation, kinetic modelling and test documentation.
- **Software:** advanced Excel (VBA, Power Query, PivotTables), Visio and Python; foundational exposure to Aspen Plus/HYSYS.

ENGINEERING EXPERIENCE

Junior Process & R&D Engineer

Mar 2021 - Jul 2022

Free Radical Process Design (FRPD) | South Africa

- Conducted laboratory investigations into industrial water-treatment and hydrometallurgical processes, including dissolved-metal removal from synthetic mine-tailings water.
- Evaluated chemical and electrochemical treatment routes; prepared mass and energy balances, process calculations, PFDs/P&IDs, design bases and reports; supported equipment sizing.
- Managed laboratory investigations, interpreted analytical results and evaluated process feasibility; prepared engineering recommendations and reviewed supplier quotations.

Master's Research in Chemical Engineering

Jan 2020 - Jun 2021

University of Pretoria | South Africa

- Designed mineral-leaching experiments and identified an internal diffusion-controlled mechanism using a D1 kinetic model; tested predictions against additional experiments and solid-residue characterisation.
- Maintained experimental records, guided research assistants and communicated findings through technical reports and peer-reviewed publication.

SELECTED WATER, ENVIRONMENTAL & R&D PROJECTS

- **Tailings-water treatment, Peru:** Tested oxidation and selective precipitation on synthetic tailings water; final analysed filtrate contained <10 mg/L each of Fe, Zn and Mn. Investigated electrolysis as a complementary route.
- **Rotating-cathode zinc removal:** Investigated electrochemical zinc removal from tailings water, including scale-up considerations for mobile-plant application.
- **Selective Zn/Mn precipitation:** Obtained final filtrate concentrations below 3.5 mg/L for each metal in preliminary laboratory tests; supported manganese sulphate monohydrate process development.
- **Fe/Al removal from leach liquor:** Investigated MgO, CaCO3 and limestone/eggshell for precipitation of dissolved iron and aluminium; assessed treatment agents for fertiliser-related processing.
- **Continuous cobalt electrowinning:** Supported development and technical evaluation of an electrochemical metal-recovery process.
- **Phlogopite leaching - MEng:** Developed and tested kinetic models using nitric-acid leaching experiments and feed/residue characterisation; findings published in Clays and Clay Minerals.
- **Additional resource-recovery work:** Tested a 2 L gold electrowinning setup over an 18-hour flow experiment; supported Ni-MH battery recovery feasibility and hydrotalcite-like material synthesis.

ADDITIONAL PROFESSIONAL EXPERIENCE

Business Setup & Marketing Consultant
Apr 2025 – Present

Cresco Consulting FZCO | UAE

- Coordinate business setup documentation, maintain records and support client communications and marketing activities.

Stewardess / Deckhand / Engineer Assistant
May 2023 - Apr 2025

Research Vessel Rosa | United States & Bahamas

- Assisted with vessel RO watermaker, pump and plumbing maintenance and troubleshooting; organised technical drawings and maintenance records and followed marine safety procedures.

Academic Journal Editor
Nov 2022 – Apr 2023

CACTUS Communications | South Africa

- Edited scientific manuscripts for technical clarity, consistency and publisher formatting requirements.

Teaching Assistant (Part-Time)
Feb 2019 – Jun 2021

University of Pretoria | South Africa

- Supported laboratory sessions and tutoring in particle technology, process kinetics and engineering practice/costing; supervised laboratory work, assessed students and provided technical feedback.

Laboratory Intern (Vacation Work)
Dec 2017 – Jan 2019

University of Pretoria | South Africa

- Completed vacation placements totalling 12 weeks in mineral leaching and fertiliser production, including experimental optimisation and analytical testing.

EDUCATION

MEng Chemical Engineering

University of Pretoria, South Africa | 2021

- Dissertation on nitric-acid leaching kinetics of phlogopite and solid-residue characterisation; awarded a postgraduate research bursary.

BEngHons Chemical Engineering (Distinction)

University of Pretoria, South Africa | 2020

- Recipient of Academic Honorary Colours.

BEng Chemical Engineering

University of Pretoria, South Africa | 2019

- Environmental Engineering specialisation (93%): water/effluent treatment, bioreactors, air pollution, waste remediation and environmental regulation.

PROFESSIONAL DEVELOPMENT

- Occupational Safety and Health: Hazard Communication – LinkedIn Learning | Feb 2026
- STCW Basic Safety Training: Fire Prevention/Firefighting, First Aid, Personal Safety & Social Responsibility | 2023
- Approved Engine Course (AEC 1) – Maritime Professional Training | 2024
- Project Management – Energy and Water Projects – PMI / LinkedIn Learning | Feb 2026
- LEED Green Associate Exam Preparation - LinkedIn Learning | Sep 2026
- Operations Management Professional Certificate – Jack Welch Management Institute / LinkedIn Learning | In Progress (2026)

LANGUAGES

English (native), Afrikaans (proficient), Hebrew (basic)